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VAEN-6.2: A Safety-Partitioned, Multi-Timescale Control Architecture for Embodied Foundation Models on Edge Silicon

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ABSTRACT

Vision-language-action (VLA) models have brought broad semantic generalization to robotic manipulation, yet their computational footprint, low inference frequency, and stochastic nature make them poorly suited as direct closed-loop controllers for dynamically unstable machines. Bridging the gap between high-latency foundation models and high-rate physical stabilization requires more than model quantization; it demands a strict architectural separation between *task capability* and *physical controllability*. We present **VAEN-6.2** (*Versatile Arbitrated Embodied Nervous-System*), a safety-partitioned, multi-timescale control architecture designed to run generalist policies on resource-constrained edge compute—specifically targeting the 16 GB NVIDIA Jetson Orin NX—without compromising bipedal balance or hardware safety. VAEN-6.2 partitions embodied intelligence across four isolated operational timescales (deliberative cognition, short-horizon state prediction, nominal trajectory generation, and real-time survival) bound together by explicit temporal freshness contracts and hardware-enforced authority limits. Under this framework, asynchronous neural policies execute on shared-memory AI accelerators and emit typed, time-bounded motion references, while a dedicated Real-Time (RT) Motion Core permanently retains ownership of state estimation, whole-body stabilization, safety projection, and joint impedance control. An independent microcontroller (MCU) supervises hardware-level invariants outside the operating system kernel. We formalize the control arbitration yoking nominal policies and fast tactile reflexes to the RT core, introduce an uncertainty-gated hybrid physics-residual latency compensator, and derive a memory-bandwidth and kernel-scheduling feasibility model for unified-memory edge SoCs. Finally, we specify a pre-registered, five-arm ablation protocol requiring every learned subsystem to demonstrate measurable closed-loop gains over analytical baselines prior to physical deployment.

Keywords: Humanoid Robotics, Vision-Language-Action (VLA) Models, Real-Time Control Architecture, Functional Safety, Edge AI Contention, Latency Compensation.

1. INTRODUCTION

The convergence of multimodal foundation models and generative action representations has reshaped robot learning. Contemporary dual-system architectures—such as Figure's Helix (Figure AI, 2025), NVIDIA's GR00T N1 series (NVIDIA, 2025), and Physical Intelligence's π_0 (Black et al., 2024)—pair a low-frequency Vision-Language Model (VLM) with a higher-frequency visuomotor policy instantiated via flow matching (Lipman et al., 2023) or diffusion heads (Chi et al., 2023). While these frameworks exhibit strong zero-shot and few-shot generalization in structured manipulation settings, deploying them onto untethered, dynamically balancing bipedal humanoids exposes fundamental systems-engineering bottlenecks that model-centric literature routinely leaves implicit.

First, legged embodiment imposes a severe **timescale and criticality mismatch**. Semantic scene interpretation can tolerate latencies of 300–500 ms, whereas bipedal balance recovery and contact-slip arrest require deterministic feedback loops operating within 1–10 ms. Second, **shared-memory resource contention** invalidates naive parameter-counting on embedded System-on-Chips (SoCs). On unified-memory architectures such as the Jetson Orin NX, the memory-bandwidth demand of streaming a multi-billion-parameter VLM during prompt prefill or token generation can starve concurrent visuomotor inference and camera ISP pipelines, inducing severe latency jitter even when all model weights fit comfortably in physical RAM. Third, **authority ambiguity** arises whenever an asynchronous action policy, a fast contact reflex, and a whole-body stabilizer attempt to modulate the same actuator degrees of freedom simultaneously. Finally, **failure semantics on a biped are non-trivial**: unlike a wheeled platform or a fixed-base manipulator where an emergency stop (`E-stop`) simply cuts motor power or engages static brakes, de-energizing the actuators of a walking humanoid mid-stride guarantees an uncontrolled mechanical collapse.

Addressing these failure modes requires treating control topology as a first-class safety constraint rather than an implementation afterthought. We propose **VAEN-6.2** (*Versatile Arbitrated Embodied Nervous-System*), an execution architecture built on the principle that high-capacity neural models should be *kinetically yoked* to a deterministic real-time spine: learned policies may scale a robot's semantic and manipulative capability, but physical controllability and balance invariants must remain strictly independent of the AI compute stack.

Rather than reporting post-hoc empirical benchmarks on a single mechanical platform, this paper presents the formal architectural specification, timing contracts, and pre-registered experimental methodology of VAEN-6.2. Our primary contributions are:

1. **Hardware and Authority Partitioning (§3 & §4):** We define a three-tier compute topology (Edge AI SoC, Real-Time Motion Core, and Independent Safety MCU) in which the RT Motion Core permanently owns state estimation, dynamic stabilization, and command arbitration. If the AI SoC suffers a process hang, thermal throttle, or deadline miss, the system transitions along a state-conditioned fallback hierarchy without relinquishing balance control.
2. **Temporal Freshness Contracts (§4.3):** We treat data staleness as an explicit control variable. Every cross-domain message carries monotonic temporal validity bounds (τ_{TTL}), preventing delayed neural outputs from silently driving physical actuators.
3. **Bounded Reflex Arbitration and Hybrid Latency Compensation (§4.4–§4.6):** We formulate a real-time projection operator that blends receding-horizon policy chunks with high-rate tactile residuals inside state-dependent authority envelopes, paired with an uncertainty-gated kinematic-residual predictor to compensate for pipeline delay.
4. **Edge Bandwidth Feasibility and Pre-Registered Ablations (§5 & §6):** We model the LPDDR5 memory-bandwidth and CUDA scheduling constraints of the 16 GB Jetson Orin NX—showing why multi-rate model specialization and bounded kernel granularity are mandatory on edge silicon—and lock in a falsifiable evaluation protocol against classical analytical baselines.

2. RELATED WORK

Generalist Policies and Dual-System Architectures. Decoupling slow deliberative reasoning from fast reactive control traces back to classic three-layer robotics architectures (Gat, 1998) and has recently re-emerged in embodied foundation models. π_0 (Black et al., 2024) couples a pre-trained VLM backbone with a flow-matching action expert to emit 50 Hz control chunks. Similarly, GR00T N1 (NVIDIA, 2025) and its subsequent N1.7 release (~3B parameters built on a Cosmos-Reason2-2B backbone) pair vision-language conditioning with diffusion-transformer action heads tailored for humanoids. Helix (Figure AI, 2025) separates a 7–9 Hz VLM from a 200 Hz reactive visuomotor policy, while SmolVLA (Shukor et al., 2025) demonstrates that sub-billion-parameter models can achieve competitive manipulation on constrained hardware. VAEN-6.2 is agnostic to the specific VLA parameterization: it treats heavy models such as GR00T as off-board teachers or high-compute profile options while enforcing the deterministic execution boundaries required to run distilled student policies safely on embedded modules.

Action Chunking and Trajectory Blending. Action Chunking with Transformers (ACT) (Zhao et al., 2023) and Diffusion Policy (Chi et al., 2023) mitigate compounding imitation errors by predicting multi-step action sequences. Because asynchronous chunk regeneration introduces boundary discontinuities when a newly inferred trajectory diverges from the active one, recent work on real-time flow execution (Black et al., 2025) addresses chunk stitching under inference latency. VAEN-6.2 adopts receding-horizon chunk execution but enforces C^2 (acceleration) and C^3 (jerk) continuity strictly inside the deterministic RT Motion Core rather than relying on generative model smoothness for hardware protection.

Legged Locomotion, Whole-Body Control, and Reflexes. Sim-to-real reinforcement learning with domain randomization (Tobin et al., 2017; Hwangbo et al., 2019) and Rapid Motor Adaptation (RMA) (Kumar et al., 2021) have become standard for robust legged locomotion, while optimization-based Whole-Body Control (WBC) (Kuindersma et al., 2016) and impedance control (Hogan, 1985) provide verifiable torque and contact constraint satisfaction. Classical analytical push-recovery theory based on the Instantaneous Capture Point (ICP) (Pratt et al., 2006) directly informs the state-dependent stopping primitives used in our fallback hierarchy (§4.7). For fast sensory adaptation, continuous-time recurrent models such as Closed-form Continuous-time (CfC) networks (Hasani et al., 2022) offer compact parameterizations with dynamic time constants. Because empirical validation of CfCs on full-scale humanoids remains limited, VAEN-6.2 isolates them within a strictly bounded reflex residual and mandates a controlled comparison against standard MLPs, GRUs, and linear filters.

Functional Safety in Dynamically Stable Robots. Traditional machinery safety standards (ISO 13849-1:2023) and personal-care robot standards (ISO 13482:2014) implicitly assume that removing actuator power brings a machine to a safe static equilibrium. Because this assumption fails for actively balanced bipeds, the emerging ISO/CD 25785-1 standard specifically addresses dynamically stable mobile robots. VAEN-6.2 aligns its fault-handling semantics with this paradigm, prioritizing controlled kinetic-energy dissipation and posture maintenance over unconditional contactor dropout.

3. PROBLEM FORMULATION AND SYSTEM REQUIREMENTS

We consider a bipedal humanoid with n actuated degrees of freedom equipped with proprioceptive sensing (IMUs, joint encoders, phase-current monitors), tactile/force-torque arrays, and RGB-D cameras. Onboard computation is partitioned across three hardware domains:

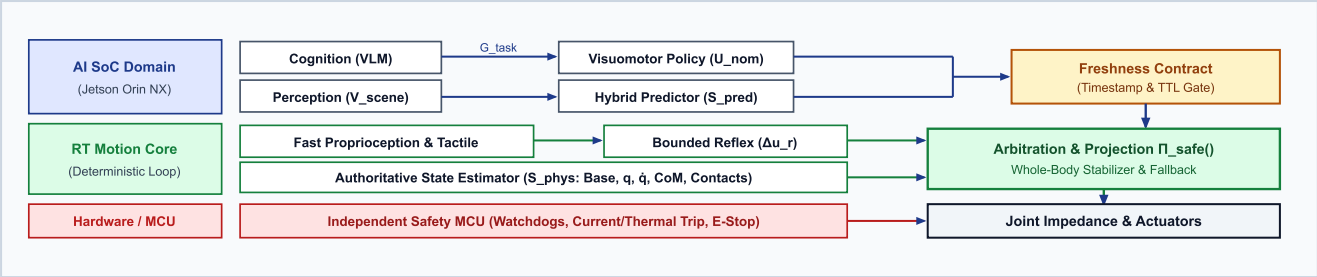
1. **The Cognitive & Policy Domain (AI SoC):** A unified-memory embedded GPU/CPU module (target baseline: NVIDIA Jetson Orin NX, 16 GB LPDDR5, 102.4 GB/s peak bandwidth) running standard Linux and neural inference runtimes.
2. **The Real-Time Motion Core (RT Core):** A deterministic processor (either an isolated RTOS/ `PREEMPT_RT` core complex or a dedicated real-time microcontroller/DSP) executing fixed-frequency control loops with bounded worst-case execution time (WCET).
3. **The Independent Safety Supervisor (Safety MCU):** A lockstep microcontroller operating without an OS, CUDA dependency, or network middleware, wired directly to driver enable lines, hardware watchdogs, and E-stop circuits.

To guarantee physical reliability despite probabilistic high-level components, the architecture enforces six core invariants. **Invariant 1 (Survivability under AI Loss)** requires that a complete hang, crash, or power loss of the AI SoC never induces a loss of dynamic balance or erratic actuator commands. **Invariant 2 (Non-Bypassable Validation)** stipulates that no learned output—whether from a VLM, action policy, or reflex network—may write directly to motor drivers or bypass RT kinematic, torque, and stability projection; best-effort GPU scheduling on the AI SoC (§5.2) is never trusted as a safety mechanism. **Invariant 3 (Explicit Staleness Rejection)** dictates that asynchronous messages lacking verifiable temporal freshness are deterministically decayed or rejected according to task criticality. **Invariant 4 (State-Dependent Fallback Semantics)** ensures that fault mitigation conditions on the robot's instantaneous kinetic energy and support polygon rather than defaulting to an immediate torque cut. **Invariant 5 (Baseline Justification)** requires every learned module in the control loop to demonstrate statistically significant closed-loop improvement over a simpler analytical or linear baseline under identical latency. Finally, **Invariant 6 (Hardware-Agnostic Hierarchy)** ensures that scaling compute capacity (e.g., migrating from an Orin NX to a Jetson Thor-class module) increases model capacity and inference rate without altering underlying safety contracts or authority boundaries.

Functional Layer	Target Execution Rate	Hardware Domain	Primary Responsibility
Cognitive Planning (VLM)	0.5–3 Hz (Event-Driven)	AI SoC (GPU)	Semantic intent parsing, task decomposition, constraint generation
Visual Perception & World Prediction	10–20 Hz	AI SoC (DLA / GPU)	Dense feature extraction, 3D object tracking, hybrid state extrapolation
Visuomotor Action Policy	25–50 Hz	AI SoC (GPU)	Receding-horizon nominal motion reference

4. THE VAEN-6.2 ARCHITECTURE

Signal flow in VAEN-6.2 is strictly hierarchical with respect to physical authority: upstream neural modules propose *motion references*, whereas downstream deterministic modules enforce *dynamic feasibility*.



4.1 The Real-Time Motion Core and Authority Arbitration

A well-documented failure mode in hybrid control architectures is the **controller switching transient**: when a robot switches from a learned policy to a backup safety controller during an emergency, structural discontinuities between the two control laws often destabilize the system faster than the external perturbation itself. VAEN-6.2 eliminates switching transients by keeping the **RT Motion Core** permanently active across all operating modes.

The RT Motion Core maintains the authoritative physical state vector $\boldsymbol{S}_{\text{phys}}(t) \in \boldsymbol{R}^{d}$—comprising floating-base pose and twist, joint positions q and velocities $\dot{q}$, estimated external wrenches, Center of Mass (CoM) state, and contact probabilities—via high-rate invariant extended Kalman filtering or factor-graph estimation. Rather than commanding raw motor currents, the upstream Visuomotor Action Policy emits a nominal reference trajectory $u_{\text{nom}}(t)$ (such as desired base velocities, footstep placements, or upper-body joint/end-effector targets). Concurrently, the fast reflex module emits a corrective residual $\Delta u_r(t)$. Inside the RT Motion Core, these signals are combined and projected onto the set of dynamically admissible commands before reaching the stabilizer and joint impedance loops:

$$u_{\text{final}}(t) = \Pi_{\text{safe}}\big(u_{\text{nom}}(t) + \text{clip}\big(\Delta u_r(t), -\boldsymbol{A}(k, \boldsymbol{S}_{\text{phys}}), +\boldsymbol{A}(k, \boldsymbol{S}_{\text{phys}}) \big) \,\big|\, \boldsymbol{S}_{\text{phys}}(t) \big) \tag{1}$$

Here, $\boldsymbol{A}(k, \boldsymbol{S}_{\text{phys}})$ is a skill- (k) and state-dependent **authority envelope** that clamps the maximum excursion a learned reflex can inject (for example, permitting rapid normal grip-force escalation during incipient object slip while restricting lower-body joint offsets that would violate capture-point margins). The projection operator $\Pi_{\text{safe}}(\cdot)$ enforces joint position, velocity, and torque limits, self-collision constraints, and whole-body balance invariants. Crucially, both $\text{clip}(\cdot)$ and $\Pi_{\text{safe}}(\cdot)$ execute exclusively within the deterministic RT address space; the AI SoC has no write access to envelope parameters.

Stabilizer Instantiation and the Agility Trade-Off. Our primary specification uses a deterministic, model-based stabilizer (Whole-Body Inverse Dynamics or Model Predictive Control) inside Π_{safe} to guarantee analytical verifiability and predictable fallback trajectories. However, classical convex stabilizers can be overly conservative during severe perturbations, restricting the rapid, non-linear angular momentum transfers that end-to-end RL policies exploit to recover from large pushes. We therefore pre-declare an architectural contingency: if model-based stabilization proves insufficiently agile during physical push-recovery trials, a compact, fixed-weight RL locomotion policy (~1–2M parameters, simple MLP) may be instantiated as the lower-body stabilizer, *provided it executes entirely inside the RT Core's deterministic memory and timing budget*, thereby preserving Invariant 1.

4.2 The Independent Safety MCU and Hardware Fault Semantics

Software assertions cannot protect against silicon lockups, bus latch-ups, or power-stage runaway. The Independent Safety MCU operates asynchronously from the main compute stack, monitoring hardware heartbeats from both the AI SoC and RT Motion Core alongside motor phase currents, MOSFET temperatures, encoder discrepancy flags, and physical E-stop loops. If the RT Motion Core itself faults or misses its watchdog window, the Safety MCU cannot compute active whole-body balance. Consequently, safe-state behavior under an RT Core failure is fundamentally an electromechanical co-design problem. Depending on the robot's kinematic state at the instant of RT failure, the MCU triggers hardware-defined mitigations inside the motor drivers: active phase-shorting (electromagnetic viscous damping) to slow gravitational collapse, engagement of passive spring/friction brakes on load-bearing knee and hip

4.3 State Representation and Temporal Freshness Contracts

Compressing high-dimensional visual scenes into an explicit geometric state vector ($\boldsymbol{S}_{\text{phys}}$) creates an information bottleneck that degrades dexterous manipulation of deformable or articulated objects. Conversely, passing unverified visual latents into a balance controller undermines safety guarantees. VAEN-6.2 maintains four distinct representations: $\boldsymbol{S}_{\text{phys}}$ (low-dimensional, metrically calibrated physical state owned by the RT Core), $\boldsymbol{V}_{\text{scene}}$ (multi-scale visual feature maps and 3D object tracklets cached on the AI SoC), $\boldsymbol{G}_{\text{task}}$ (structured semantic intent and constraints emitted by the VLM), and $\boldsymbol{S}_{\text{pred}}$ (short-horizon predicted physical state compensating for pipeline delay).

To prevent silent consumption of stale data across asynchronous clock domains, every message header m crossing an interface boundary carries a synchronization tuple $\langle t_{\text{stamp}}, \text{seq_id}, \text{frame_id}, t_{\text{valid_until}} \rangle$ referenced to a shared monotonic hardware clock (or Precision Time Protocol [PTP] across physical buses). Upon receipt at time t_{now}, the consumer computes packet age $a = t_{\text{now}} - t_{\text{stamp}}$ and evaluates it against a skill-specific Time-To-Live threshold $\tau_{\text{TTL}}(k)$:

$$\text{Status}(m) = \begin{cases} \textbf{VALID}, & \text{if } a \le \tau_{\text{TTL}}(k) \text{ and } t_{\text{now}} \le t_{\text{valid_until}} \\ \textbf{DECAY}, & \text{if } \tau_{\text{TTL}}(k) < a \le \tau_{\text{TTL}}(k) + \Delta_{\text{ext}}(k) \\ \textbf{EXPIRED}, & \text{otherwise.} \end{cases} \tag{2}$$

A single global timeout is physically inappropriate: quasi-static stance or slow walking can tolerate a 100 ms visual feature age, whereas dynamic grasping or slip recovery requires $\tau_{\text{TTL}} \le 15$ ms. While a packet is in the `DECAY` window $\Delta_{\text{ext}}(k)$, the RT Core smoothly attenuates command velocity toward zero; once it reaches `EXPIRED`, the stale reference is stripped and the local state machine transitions to its degraded fallback mode (§4.7).

4.4 Hybrid Physics-Residual Latency Compensation

In an embodied VLA pipeline, the total closed-loop delay L_{total} from sensor transduction to mechanical torque generation is the sum of sequential processing and queuing delays:

$$L_{\text{total}} = L_{\text{sensor}} + L_{\text{transport}} + L_{\text{perception}} + L_{\text{estimator}} + L_{\text{policy}} + L_{\text{RT}} + L_{\text{actuation}} \tag{3}$$

On embedded compute, L_{total} routinely spans 40–120 ms. Supplying a policy with delayed state $\boldsymbol{S}_{\text{phys}}(t - L_{\text{total}})$ during rapid motion induces phase lag, limit-cycle oscillations, and contact overshoot. Rather than relying on a heavy generative world model to hallucinate future video frames, we restrict prediction to the low-dimensional physical state over the measured delay horizon $\Delta \approx L_{\text{total}}$ via a **hybrid physics-residual formulation**:

$$\hat{s}(t + \Delta) = f_{\text{phys}}(s(t), u_{t-\Delta\,:\,t}) + w(\sigma) \cdot g_\theta(s(t), u_{t-\Delta\,:\,t}, c(t)) \tag{4}$$

Here, $f_{\text{phys}}(\cdot)$ is a zero-parameter analytical forward integration of rigid-body kinematics and known actuator delay dynamics over the ring buffer of recently dispatched commands $u_{t-\Delta\,:\,t}$. The lightweight network $g_\theta(\cdot)$ predicts a state residual conditioned on the estimated discrete contact mode $c(t)$ (e.g., rigid stance, free swing, incipient slip, impact) to capture unmodeled transmission compliance, backlash, and surface deformation.

Handling Contact Discontinuities. Extrapolating state across rigid contact transitions is notoriously ill-conditioned; predicting a foot strike 20 ms early can cause the controller to apply stance torques in mid-air. Consequently, g_θ outputs an aleatoric uncertainty estimate $\sigma(t)$. A monotonic gating function $w(\sigma) \in [0, 1]$ attenuates the learned residual toward zero as uncertainty rises near contact transitions, smoothly reverting to conservative kinematic extrapolation or shortening the effective horizon Δ.

4.5 Action Policy and Receding-Horizon Chunk Execution

The visuomotor action policy consumes $\{\boldsymbol{G}_{\text{task}}, \boldsymbol{V}_{\text{scene}}, \boldsymbol{S}_{\text{phys}}, \hat{s}(t+\Delta)\}$ and emits a nominal action chunk $U_{\text{nom}} = \{u_{\text{nom}}(\tau)\}_{\tau=t}^{t+T_h}$ over a prediction horizon $T_h \approx 250$ ms, regenerated every execution window $T_e \approx 40$ ms (25 Hz). To balance generalist flexibility with safety verification, VAEN-6.2 supports a dual-pathway skill interface: (1) **Structured Skill Primitives**, in which the policy outputs parameterized references for strongly typed skills (`LOCOMOTE_VEL`, `STEP_TARGET`, `EE_IMPEDANCE_REACH`, `GRASP_FORCE`) with tightly pre-verified safety invariants and τ_{TTL} bounds; and (2) **Free-Form Whole-Body References**, in which the policy emits dense joint- and task-space trajectories for long-tail tasks (such as wiping contoured surfaces or whole-body bracing). Regardless of pathway, the RT Motion Core never executes raw chunk boundaries directly; incoming chunks are timestamp-aligned and blended into the active trajectory over a cross-fade window using quintic spline interpolation that enforces maximum acceleration ($\ddot{q}_{\max}$) and jerk ($q^{\dddot{}}_{\max}$) bounds.

4.6 Bounded Tactile and Proprioceptive Reflexes

While modern RL locomotion policies internalize low-frequency push recovery, high-frequency contact phenomena in manipulation—such as incipient object slip, unexpected obstacle contact, or brittle surface impact—unfold over 4–10 ms, far faster than a 25–50 Hz vision-conditioned policy can react. The Bounded Reflex module runs at 100–250 Hz, mapping wrist force/torque, tactile taxel arrays, and joint torque-error residuals ($\tau_{\text{meas}} - \tau_{\text{model}}$) directly to Δu_r. Because this loop bypasses visual perception, it is intentionally local and myopic: it can stiffen local joint impedance, scale normal grasp force upon shear detection, or retract an end-effector along a contact normal, while its global authority remains strictly clamped by $A(k, \boldsymbol{S}_{\text{phys}})$ in Equation (1) so it cannot perturb whole-body balance.

4.7 Event-Driven Cognition and Graceful Degradation Hierarchy

Executing a 2B–3B parameter VLM continuously on an edge SoC saturates memory bandwidth and thermal headroom. In VAEN-6.2, the cognitive VLM operates asynchronously on an **event-driven** basis, remaining idle until woken by a lightweight Event Manager. We explicitly avoid raw pixel-entropy or frame-differencing triggers, as bipedal head sway and illumination shifts alter pixel intensities continuously during normal walking. Instead, VLM inference is escalated by four structured conditions: (i) *kinematic/dynamic prediction surprise*, where $\|\boldsymbol{S}_{\text{phys}}(t) - \hat{s}(t)\| > \varepsilon_{\text{surprise}}$ (e.g., an object is much heavier than expected or a cabinet door is jammed); (ii) *semantic tracker novelty*, triggered by the appearance or loss of tracked 3D object tracklets in $\boldsymbol{V}_{\text{scene}}$; (iii) *skill terminal conditions or timeouts* of the active $\boldsymbol{G}_{\text{task}}$ primitive; and (iv) a *low-rate background sweep* (~0.5 Hz) to capture gradual environmental drift.

Deterministic Degradation State Machine. System health is governed by a finite-state machine maintained inside the RT Motion Core:

```
FULL_OPERATION → COGNITIVE_DEGRADED → PREDICTION_DEGRADED → VISION_DEGRADED → POLICY_DEGRADED
                        → AI_OFFLINE → RT_FAULT → HARDWARE_EMERGENCY
```

Here, `POLICY_DEGRADED` is explicitly distinguished from `AI_OFFLINE`: it captures scenarios where the AI SoC and VLM remain responsive over the bus, but the visuomotor policy process misses its deadline, emits malformed packets, or stalls internally. The RT Core treats `POLICY_DEGRADED` and `AI_OFFLINE` identically once U_{nom} ages past $\tau_{\text{TTL}} + \Delta_{\text{ext}}$, transitioning immediately to a **state-conditioned survival policy** based on instantaneous kinetic energy:

- *Quasi-static double support ($\|\dot{x}_{\text{CoM}}\| < \varepsilon_v$):* Freeze the current CoM projection inside the support polygon and hold nominal standing joint impedance.
- *Dynamic locomotion:* Execute an Instantaneous Capture Point (ICP) deceleration sequence (Pratt et al., 2006)—a 1–2 step stopping primitive computed directly from $\boldsymbol{S}_{\text{phys}}$—to bring horizontal CoM velocity to zero before locking standing posture.
- *Unrecoverable state (CoM outside capture basin):* Retract extremities, lower the CoM, and switch joint impedance to high-damping/low-stiffness compliance to dissipate impact energy and protect drivetrain gearboxes.

5. EDGE DEPLOYMENT FEASIBILITY AND CONTENTION ANALYSIS

Equating memory capacity with deployment feasibility is a common pitfall in edge robotics. Showing that a quantized 3B-parameter VLA fits within 16 GB of RAM provides no guarantee that it can execute alongside high-rate perception and control pipelines without inducing severe latency jitter.

5.1 Memory Bandwidth as the Primary Bottleneck

The NVIDIA Jetson Orin NX (16 GB) features a 128-bit LPDDR5 memory subsystem with a theoretical peak bandwidth of B_{peak} = 102.4 GB/s shared across its 8-core ARM Cortex-A78AE CPU, 1024-core Ampere GPU, two Deep Learning Accelerators (DLAs), and camera ISP/DMA engines. Under realistic mixed read/write workloads, sustainable bus throughput is approximately $\eta B_{\text{peak}} \approx$ 75–85 GB/s. For memory-bandwidth-bound neural inference, streaming an active weight footprint of W_i gigabytes at frequency F_i Hz requires a minimum bandwidth of $B_i \approx W_i \cdot F_i$. Across M concurrent workloads, system feasibility requires:

$$\sum_{i=1}^{M} W_i F_i + B_{\text{ISP/DMA}} + B_{\text{OS/CPU}} \leq \eta B_{\text{peak}} \tag{5}$$

Equation (5) immediately rules out monolithic high-rate architectures on edge silicon: running a 2 GB policy network at 200 Hz would demand 400 GB/s of memory traffic—nearly five times the physical ceiling of the Orin NX. Even running a 1.5 GB action expert at 50 Hz consumes 75 GB/s, starving the CPU, OS, and camera DMA channels.

Subsystem Module	Param Budget	Precision	Footprint (W_i)	Rate (F_i)	Bandwidth (W_iF_i)	Compute Assignment
Fast Visual Backbone	15–30 M	FP16 / INT8	~0.04 GB	20 Hz	~0.8 GB/s	DLA 0 / GPU
Semantic Vision Encoder	80–150 M	INT8	~0.15 GB	10 Hz	~1.5 GB/s	DLA 1 / GPU
Hybrid State Predictor (g_θ)	5–15 M	FP16	~0.03 GB	20 Hz	~0.6 GB/s	GPU (High-Priority Stream)
Visuomotor Action Policy	40–80 M	FP16	~0.15 GB	30 Hz	~4.5 GB/s	GPU (High-Priority Stream)
Bounded Reflex (Δu_r)	1–5 M	FP16	~0.01 GB	200 Hz	~2.0 GB/s (target: cache-resident)	Isolated CPU Core / RT Core
Cognitive VLM (Quantized)	1.5–3.0 B	INT4 (AWQ)	~1.80 GB	Event (≤2 Hz)	~3.6–18 GB/s (Burst)	GPU (Low-Priority Chunked)
OS, Camera ISP, Buffers	—	—	~4.50 GB	Continuous	~8.0 GB/s	CPU / NVCSI / ISP
Total Steady-State / Peak	**—**	**—**	**~6.68 GB**	**—**	**17.4 GB/s (Steady) / 35.4 GB/s (Burst)**	**< 45% of ~80 GB/s practical limit**

Table 2: Analytical memory footprint and bandwidth budget on a 16 GB Jetson Orin NX. Steady-state bandwidth (17.4 GB/s) excludes the event-driven VLM; peak burst (35.4 GB/s) includes worst-case VLM token streaming. The "cache-resident" target for the reflex model cannot be assumed a priori — the Cortex-A78AE CPU complex provides only 2 MB of L2 and 4 MB of shared system L3 cache, and actual residency under concurrent memory-bus traffic from the vision and VLM workloads must be verified by hardware profiling rather than assumed from parameter count alone.

5.2 Mitigating VLM Prefill Contention and Scheduling Jitter

While steady-state operation consumes only ~17.4 GB/s, the primary timing hazard on the Orin NX is **VLM prefill contention**: when the Event Manager wakes the 3B VLM, processing visual and text prompt tokens generates an intense compute and memory-bus burst that can delay a pending 30 Hz Action Policy kernel launch by tens of milliseconds. To bound this interference on shared silicon, VAEN-6.2 combines three isolation mechanisms:

1. **Hardware Engine Offloading:** Static-graph vision backbones are compiled onto the Orin's dedicated Deep

6. PRE-REGISTERED EVALUATION PROTOCOL AND FALSIFIABLE HYPOTHESES

In multi-layered robotics architectures, overall task success is easily misattributed to complex learned modules when closed-loop stability is actually carried by the underlying PD impedance controller. To prevent post-hoc rationalization, we pre-register three falsifiable hypotheses and their corresponding ablation arms prior to physical trials. Any learned module that fails to outperform its simpler analytical baseline under matched latency will be pruned from the stack.

6.1 Core Hypotheses and Ablation Arms

> **Hypothesis 1 (Closed-Loop Utility of the Bounded Tactile Reflex):** *During contact-rich manipulation subjected to unmodeled friction shifts and external force impulses, adding the* 200 Hz *bounded reflex residual (Δu_r) significantly improves slip-arrest success rate and reduces peak contact-force overshoot relative to the* 30 Hz *visuomotor policy and passive joint impedance alone.*
> **Ablation Arms:** `(R0)` No reflex (policy + RT impedance only); `(R1)` Classical high-pass force-derivative filter with linear impedance scaling; `(R2)` 3-layer MLP reflex; `(R3)` GRU recurrent reflex; `(R4)` Closed-form Continuous-time (CfC) network. All neural arms (`R2` – `R4`) are restricted to an identical 2M-parameter budget, identical sensory inputs, and matched training rollouts.

> **Hypothesis 2 (Closed-Loop Utility of Hybrid Physics-Residual Prediction):** *Under realistic end-to-end pipeline delays ($L_{\text{total}} \in [40, 100]$ ms), conditioning the action policy on the hybrid predicted state $\hat{s}(t + \Delta)$ yields lower trajectory tracking error and higher dynamic manipulation/locomotion success than uncompensated state or purely analytical extrapolation.*
> **Ablation Arms:** `(P0)` No prediction ($\boldsymbol{S}_{\text{phys}}(t)$ passed directly); `(P1)` Constant-velocity kinematic extrapolation; `(P2)` Analytical rigid-body dynamics integration (f_{phys} only); `(P3)` Pure end-to-end learned state predictor (g_θ only); `(P4)` Full uncertainty-gated Hybrid Predictor ($f_{\text{phys}} + w(\sigma)g_\theta$).

> **Hypothesis 3 (Efficiency of Event-Driven Cognition):** *Event-driven VLM escalation triggered by prediction surprise and semantic novelty reduces cumulative GPU energy consumption and 99th-percentile (p99) policy inference jitter by at least* 50% *compared to fixed-rate (*2 Hz*) VLM polling, while maintaining task completion rates within a* 5% *equivalence margin.*

6.2 Staged Verification Pipeline

Experimental validation progresses through four sequential gates:

1. **Delay-Injected Physics Simulation:** Policies and predictors are trained and benchmarked in simulation (Isaac Lab / MuJoCo) under realistic transport delays, timestamp jitter, dropped camera frames, and state-estimator bias rather than ground-truth state.
2. **Deterministic On-Target Log Replay:** Recorded multisensor streams from physical hardware—preserving exact monotonic timestamps and sequence IDs—are replayed on the 16 GB Orin NX to measure deadline-miss rates, p99 policy latency, and thermal throttling during concurrent VLM prefill bursts.
3. **Hardware-in-the-Loop (HIL) Fault Injection:** With the robot suspended in a safety gantry, we inject systematic faults—abrupt termination (`SIGKILL`) of the AI SoC policy process, out-of-bounds reference corruption, simulated encoder dropout, and artificial packet expiration ($a > \tau_{\text{TTL}}$)—verifying that the RT Motion Core transitions smoothly to its state-dependent fallback without torque discontinuities.
4. **Physical Disturbance Trials:** Multi-trial physical evaluations reporting 95% confidence intervals across standardized push-recovery impulses, unannounced payload mass changes, and low-friction grasping benchmarks.

7. LIMITATIONS, STRUCTURAL TRADE-OFFS, AND OPEN FAILURE MODES

Prioritizing determinism, edge deployability, and hard safety boundaries introduces fundamental engineering trade-offs that constrain VAEN-6.2:

The Conservatism–Agility Bottleneck. Enforcing Π_{safe} and clamping reflex authority via $\boldsymbol{A}(k, \boldsymbol{S}_{\text{phys}})$ guarantees

Upfront Systems Engineering Overhead. Implementing a hard-real-time state estimator, whole-body stabilizer, versioned IPC contract layer, and lockstep Safety MCU firmware requires substantial core robotics engineering before any learned policy can be tested. While unnecessary for stationary tabletop arms, this overhead is dictated by the kinetic energy and fall risks of untethered bipedal humanoids.

Distillation Capacity Ceiling. Compressing a 3B+ generalist VLA into a 40–80M parameter student policy for 30 Hz edge execution inevitably trades open-vocabulary, zero-shot generalization on novel object categories for closed-loop reliability across structured skill families.

Dual AI and RT Core Failure. If both the AI SoC and the RT Motion Core fail simultaneously, only the Independent Safety MCU remains in the loop (§4.2). Because the MCU cannot compute active balance control, its ability to mitigate an uncontrolled fall is strictly bounded by passive electromechanical properties selected at hardware design time (joint back-drivability, phase-shorting damping, and mechanical brakes), not by software architecture.

8. CONCLUSION

As embodied AI moves from cloud-tethered laboratory demonstrations to untethered bipedal machines operating alongside humans, the primary engineering bottleneck shifts from *model representation capacity* to *closed-loop systems predictability*. VAEN-6.2 provides a concrete architectural blueprint for deploying probabilistic foundation models on bandwidth-constrained edge silicon without surrendering physical control to a non-deterministic black box. By yoking high-level neural policies to a permanent real-time stabilization spine, elevating data staleness to an explicit control variable, and requiring every learned module to outperform analytical baselines under realistic latency, this work establishes a falsifiable foundation for safe humanoid autonomy. Subsequent work will report empirical ablation results from our hardware-in-the-loop testbed and physical bipedal platform.

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